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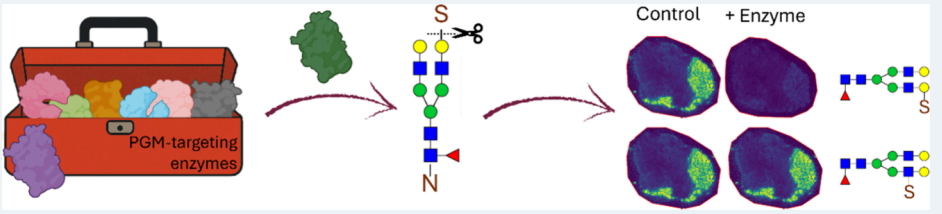
Application of Post-Glycosylation Modifying Enzymes for Mass Spectrometry Imaging of Modified *N*-Glycans *In Situ*

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ABSTRACT: Glycans are essential components of cells involved in numerous biological processes, and changes in glycan profiles are often correlated with disease progression. Glycans are comprised of various monosaccharides linked together at different positions with varied stereochemistry. The structural diversity and complexity present unique analytical challenges that can limit understanding of their functional role. In addition, glycans are frequently decorated with a diverse set of chemical modifications, termed post-glycosylation modifications (PGMs). Characterization of PGMs is essential for a thorough understanding of the glycome; however, the technical challenges and low throughput of current methodologies limit our understanding of PGMs. Here, we demonstrate a novel approach for rapid visualization of specific PGMs present in tissue *N*-glycans by applying PGM-targeting enzymes to matrix-assisted laser desorption ionization-based mass spectrometry imaging. The method enables *in situ* investigation of PGMs, allowing identification of the modified sugar residue, while visualizing the spatial distribution of each modified *N*-glycan. As the repertoire of PGM-targeting enzymes expands, we anticipate that this approach will improve our understanding of PGM distributions within the dynamic *N*-glycome, providing new biological insights into the identification of novel disease biomarkers.

KEYWORDS: *postglycosylation modification, mass spectrometry imaging, modified N-glycans, biomarker screening*

INTRODUCTION

Glycans are saccharide-based polymers essential to countless biological and pathological processes encompassing structural functions as well as modulation of intrinsic and extrinsic interactions, including cell signaling, receptor ligand binding, protein folding and stability, and surface adhesion.[1] Glycan functions are determined by their composition and structure, which is in turn defined by the monosaccharides that constitute them, their order of arrangement, and the linkages between units.

Glycan complexity is further heightened by chemical modifications to individual monosaccharides, termed post-glycosylation modifications (PGMs). PGMs encompass a range of diverse chemical groups, including sulfate, phosphate, acetyl, methyl, phosphorylcholine, aminoethylphosphonate, and phosphoethanolamine.[2−5] The wide diversity in glycan PGMs, including the monosaccharides modified, the carbon positions occupied, and the distinct chemistry of each, alters how glycans interact with surrounding molecules. Numerous diseases are associated with changes in PGMs, including autoimmune disorders, several cancers, cystic fibrosis, and congenital disorders.[4] In these studies, changes to PGMs are measured by mass spectrometry-based techniques, the use of monoclonal antibodies, or indirectly, through analysis of transcription rates of the transferases and hydrolases responsible for the modification. PGMs represent a promising avenue of study for therapeutic and diagnostic discovery. Unfortunately, current methodologies make it difficult to identify and pinpoint PGMs, thereby limiting our understanding of their distribution, and how PGM patterns correspond to disease states.

Existing glycan analysis methods that identify both the modified monosaccharide and the carbon-position of the PGM are laborious and technically challenging. Common methods used to profile and characterize glycan mixtures involve mass

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spectrometry (MS), such as liquid chromatography coupled with tandem MS (LC-MS/MS) and matrix-assisted laser desorption ionization-time-of-flight (MALDI-TOF). While the mass measurement of a glycan may suggest the presence of a PGM, it does not indicate the specific sugar residue or the carbon to which it is attached. Additional methods are then necessary to fully investigate these samples, as glycans frequently contain isomeric PGMs.[6,7] Tandem and multistage MS (MS/MS, MS^n) and NMR are used to fully characterize PGM positioning on glycans, however, these methods are technically challenging and are limited to the analysis of a single glycan at a time, often with a requirement for large inputs of purified sample.[7,8] Moreover, ionization artifacts complicate the analysis of fragile PGM-containing glycans, which may undergo in-source fragmentation or rearrangements, complicating confident structural assignment.[9,10] Differential ionization efficiency and ion suppression affect the limits of detection of particularly low-abundance or multiply modified glycoforms, making them less amenable to *in situ* analyses of cellular or tissue surfaces.[11]

Mass spectrometry imaging (MSI) is revolutionizing spatial biology by enabling the precise localization of specific chemical signatures.[12] In recent years, the endoglycosidase PNGase F has been applied to matrix-assisted laser desorption/ionization mass spectrometry imaging (MALDI-MSI) for rapid spatial analysis of *N*-glycans in sectioned tissues.[13,14] This approach provides an *in situ* snapshot of the structures and relative abundance of *N*-glycans, as well as their localization and distribution throughout the tissue.[15] When superimposed with tissue histopathology analyses, MSI can both spatially map the *N*-glycome and visualize how it may change between healthy and diseased tissues. Since PGMs can vary across cell types and tissue microenvironments, spatially resolved methods are essential for understanding their biological roles and diagnostic potential. Such a holistic view of the dynamic *N*-glycome will better inform both basic biology and enable biomarker discovery. However, MSI exploration of PGMs in *N*-glycome remains technically complicated. Recent MSI studies have identified potential sulfated *N*-glycans, but individual MS/MS validation of each structure was impractical for every observed modified *N*-glycan.[16,17]

To address the technical challenges of PGM identification in MSI, we used a suite of highly specific enzymes targeting distinct classes of PGMs to enable rapid *in situ* identification. We demonstrate the use of a phosphorylcholine-cleaving enzyme that targets phosphorylcholine *N*-glycan modifications to visualize their spatial distribution in transverse sections of a parasitic nematode using MSI. We then illustrate the use of a sulfate-specific glycosyl hydrolase, as well as several sulfatases with different specificities, to map mammalian *N*-glycan sulfation using MSI. We also present identification of new sulfatases that remove sulfate from the 3-carbon or 6-carbon of terminal galactose residues, a novel specificity. Our study validates the use of PGM-targeting enzymes to spatially visualize different PGMs using MSI and further expands the repertoire of analytical enzymes for emerging MSI workflows.

METHODS

Protein Expression and Purification

The GlcNAc-6-phosphodiesterase (PDase) and GlcNAc-6-SO_4-hexosaminidase (F10-ORF19p) were purified as previously described.[2,18] The exogalactose-3-sulfatase (BT1636p), exogalactose-6-sulfatase (BT3109p), endogalactose-3-sulfatase (BT4683p), exogalactose-3/6-sulfatase (N4-ORF16p), and exogalactose-3/6-sulfatase (E8-ORF27p) were cloned into the vector pJS119K with a C-terminal 6xHis tag using the NEBuilder HiFi DNA Assembly Cloning kit (NEB, E5520).[19] Primers were designed using the NEBuilder online resource (https://nebuilder.neb.com) (Table S1). The genes for BT1636, BT4683, and BT3109 were cloned without the DNA encoding their predicted signal peptide sequences (residues 2−21, 2−21, and 2−17, respectively). Site-directed mutagenesis of BT3109p was performed using the Q5 Site-Directed Mutagenesis Kit (NEB, E0554S) was used to convert the catalytic serine to a cysteine (S57C), enabling conversion to formyl glycine in *E. coli* as previously described.[20] The resulting plasmids were used to transform NEBExpress Competent *E. coli* cells (NEB, C2523). Cultures were inoculated in 1 L of LB containing 50 μg/mL kanamycin (ThermoFisher, J17924−06) and grown at 37 °C with shaking at 225 rpm. Once the culture reached exponential phase ($OD_{600\ nm}$ = 0.4−0.6), protein expression was induced with 0.4 mM IPTG (Teknova, I3502). Induction was performed overnight at 37 °C for N4-ORF16p, BT3109p, and BT4683p. Induction was performed overnight at 30 °C for BT1636p and E8-ORF27p. Cells were harvested and resuspended in binding buffer (20 mM sodium phosphate, 300 mM NaCl, 20 mM imidazole, pH 7.4). Lysis was performed using a Shearjet HPL60 High Pressure Homogenizer (Maximator, Germany) with three passes at 18 kPsi. Proteins were purified on a HisTrap Fast Flow column (Cytiva, 17525501). Protein was eluted from the column with 500 mM imidazole. The resulting proteins were buffer exchanged into an imidazole-free buffer (20 mM sodium phosphate, 300 mM NaCl, 1 mM EDTA, pH 7.4) using Zeba Spin Desalting Columns, 7 kDa MWCO (ThermoFisher, 89894).

Sample Preparation for MALDI-MSI

Ascaris suum. Adult *A. suum* nematode specimens were sourced from pigs, washed, and frozen at −80 °C. Portions of female worms (1 cm) were mounted using a drop of Neg-50 frozen section medium (Epredia, 6502), and transverse sections were cut at −20 °C on a Leica GM1860 cryostat (Leica Microsystems GmbH, Wetzlar, Germany) at a thickness of 12 μm. Sectioning was performed using Cryofilm type 3C(16UF) tape, and the tape containing the section was affixed with SCMM-MS1 mounting medium (SECTION-LAB Co. Ltd., Yokohama Kanagawa, Japan).[21] Samples were subjected to washes with ethanol and Carnoy's solution (10% glacial acetic acid, 30% chloroform, 60% ethanol) followed by antigen retrieval in 10 mM citraconic anhydride (TCI, M0365) buffer, pH 3 for 60 min in a 60 °C water bath as previously described.[14,22] The samples were then dried in a desiccator. A hydrophobic barrier was drawn (Vector Laboratories, H-4000) around the section and the PDase was spotted onto the samples at 0.1 mg/mL enzyme in 50 mM Tris, pH 8.0, 5 mM $MgCl_2$ and incubated at 37 °C for 16 h in a humidity chamber.[23] The concentration used was based on the concentrations of on-tissue enzymatic digestions performed in previous studies.[14] Slides were then rinsed sequentially in the following solutions to remove excess enzymes and salt: 1 min in 100% ethanol, 95% ethanol, 70% ethanol, and HPLC water; 3 min in 20 mM Tris, pH 9.0; 1 min in HPLC water; 3 min in 10 mM citraconic anhydride buffer (TCI, M0365), pH 3; 1 min in HPLC water. The samples were then dried in a desiccator.

Human Serum. Human serum (Sigma, P2918) was mixed 2:1 with 100 mM sodium bicarbonate, pH 8.0, and spotted onto an amine-reactive slide (Grace Biolabs, 246865), using a 64-well ProPlate Slide Module to guide spot placement (Grace Biolabs, 63484-48).[24] Spots were immobilized on the slide by immediately placing the slide in a preheated humidity chamber (37 °C) and incubating at room temperature for 1 h. The slide was then placed in a desiccator for 15 min to dry the spots. To desalt the samples, the spots were incubated in Carnoy's solution (10% glacial acetic acid, 30% chloroform, 60% ethanol) for 3 min, three times. After rinsing with HPLC-grade water, the samples were dried in a desiccator.

For stepwise exoglycosidase digestions, the ProPlate Slide Module was kept in place to contain the reactions. α2-3,6,8,9-neuraminidase A (NEB, P0722) was applied to the sample at 0.1 mg/mL, β1-3,4-

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galactosidase (NEB, P0746) was applied at 0.1 mg/mL, and the GlcNAc-6-SO_4 hexosaminidase (F10-ORF19p) at 6 μg/mL. The concentrations used for the MSI enzyme incubations (0.1 mg/mL) were based on the concentrations of on-tissue enzymatic digestions performed in previous studies.[14] The concentration of F10-ORF19p used was optimized in-solution to prevent off-target activity we observed on unmodified GlcNAc residues (Figure S1). Enzymatic reactions were incubated at 37 °C for 16 h in a humidity chamber,[23] and samples were rinsed with water between reactions. Slides were then rinsed sequentially in the following solutions to remove excess enzymes and salt: 1 min in 100% ethanol, 95% ethanol, 70% ethanol, and HPLC water; 3 min in 20 mM Tris, pH 9.0; 1 min in HPLC water; 3 min in 10 mM citraconic anhydride buffer, pH 3; 1 min in HPLC water. The samples were then dried in a desiccator.

Thyroid Tissue. Fresh frozen cancerous human thyroid tissue (BioIVT) was mounted using a drop of Neg-50 frozen section medium (Epredia, 6502)) and sectioned at −16 °C on a Leica GM1860 cryostat (Leica Microsystems GmbH) at a thickness of 12 μm. Sections were thaw-mounted onto SuperFrost Plus adhesion microscope slides (Epredia) and placed in a desiccator to dry. One of the sections was stained with hematoxylin and eosin (H&E) histological stain to visualize tissue pathology by incubating in the following solutions: 30 s in 95% ethanol, 30 s in 70% ethanol, 30 s in HPLC water, 2 min in Hematoxylin, 30 s in HPLC water, 30 s in 70% ethanol, 30 s in 95% ethanol, 1 min in Eosin-Y solution, 30 s in 95% ethanol, 30 s in 100% ethanol, and 2.5 min in xylenes. High-resolution images of the H&E-stained tissue were captured using a Hamamatsu NanoZoomer SQ System at 20× magnification and subsequently exported at 5× magnification (Hamamatsu). Samples were subjected to washes with ethanol and Carnoy's solution (10% glacial acetic acid, 30% chloroform, 60% ethanol), followed by antigen retrieval in 10 mM citraconic anhydride buffer (TCI, M0365), pH 3, for 5 min in a NxGen Decloaking Chamber (Biocare Medical) at 110 °C as previously described.[14,22] The samples were then dried in a desiccator. A hydrophobic pen was used to draw a barrier around each thaw-mounted sample to contain the spotted on sulfatases. Sulfatase reactions were performed with 0.1 mg/mL enzyme in 50 mM Tris, pH 8.0, at 37 °C for 16 h in a humidity chamber,[23] using 5 mM $CaCl_2$ for BT1636p or 5 mM $MgCl_2$ for N4-ORF16p, respectively. Slides were rinsed to remove the enzyme and salts as described above for the serum sample, then dried in a desiccator.

Sample Preparation for MALDI-MSI. Glycerol-free concentrated PNGase F (0.1 mg/mL, formulated internally) was applied to samples using an HTX M3+ (HTX Technologies LLC) using 15 passes with a nozzle temperature of 45 °C, a nozzle height of 40 mm, a nozzle velocity of 1200 mm/min, a flow rate of 25 μL/min, a track spacing of 3 mm, and a nitrogen pressure of 10 psi. Following application, the samples were incubated at 37 °C in a humidity chamber[23] for 2 h and then dried in a desiccator for 10 min.

Samples were coated with α-cyano-4-hydroxycinnamic acid (CHCA) (Bruker, 8201344) using an HTX M3+ sprayer (HTX Technologies LLC). A solution of 10 mg/mL CHCA in 70% acetonitrile, 0.1% trifluoroacetic acid, 20 mM ammonium phosphate was applied in three passes with a nozzle temperature of 75 °C, a nozzle height of 40 mm, a nozzle velocity of 1200 mm/min, a flow rate of 120 μL/min, a track spacing of 3 mm, and a nitrogen pressure of 10 psi.

MALDI-MSI Data Acquisition and Analysis

Images were acquired on a Bruker timsTOF fleX mass spectrometer (Bruker Daltonics) in positive mode at a spatial resolution of 50 μm over an m/z 700−4000 mass range with a Funnel 1 RF of 500 Vpp, a Funnel 2 RF of 500 Vpp, a Multiple RF of 5400 Vpp, a Collision Energy of 15 eV, a Collision RF of 4000 Vpp, a Transfer Time of 110 μs, a PrePulse Storage Time of 24 μs, and a laser power of 35%. A total of 600 laser shots were summed per pixel at 10 kHz. These settings were optimized based on signal intensity over the range m/z 700−4000 range, with an aim to maximize signal/noise. MALDI calibration was performed using red phosphorus (Sigma 343242) from m/z 774.3435−3995.6147.

Data were analyzed using the SCiLS Lab software (Bruker, version 2025b 13.01.17192), the spectra were root mean squared normalized, and hot spot removal was performed. *N*-glycan peaks were manually identified through comparison to previously characterized *N*-glycans.[6,14,25,26] All experiments were performed in duplicate.

Functional Metagenomic Screening for Sulfate-Targeting Enzymes

4-Methylumbelliferyl (4MU) galactose-3-SO_4 (Glycouniverse, GU-CSYN-046) was used in a coupled fluorescence assay to screen for novel galactose-3-sulfatases as previously described.[2] Briefly, clones from a human gut microbiome metagenomic fosmid library were grown statically overnight at 37 °C in 384-well plates containing 50 μL of Lennox Broth supplemented with 12.5 μg/mL chloramphenicol (Thermofisher, B20841.14) and 1X autoinducing solution (Lucigen). To perform the screen, 50 μL of Y-PER lysis buffer (Thermo Scientific, 78990) supplemented with 120 μM of 4MU-Gal-3-SO_4 and 1 U/mL of β1−3 Galactosidase (NEB, P0726) was added to each well. The plates were then incubated statically at 37 °C for 48 h. Fluorescence was read at λ_{ex} = 350 nm and λ_{em} = 445 nm after 1, 6, 24, and 48 h using a SpectraMax Plus Microplate Reader (Molecular Devices). Clones were identified as hits if they exhibited fluorescence at least three standard deviations above the mean background fluorescence (excluding outliers). Fosmids from hit clones were sequenced on an RSII sequencer (Pacific Bioscience). The assembled nucleotide sequences of the 5 fosmid inserts (NEB131-E8, NEB132-J4, NEB133-N4, NEB139-F8, & NEB132-F12) were deposited in GenBank under the accession numbers PV807771-5, respectively. ORFs were predicted with MetaGeneMark[27] and annotation of the predicted ORFs was performed using an accelerated profile HMM search.[28]

Novel Sulfatase Characterization

Sulfatase Activity on Monosaccharides. To identify the metal ion cofactor requirement for N4-ORF16p and E8-ORF27p, 1 μM enzyme was incubated with 30 μM 4MU-Gal-3-SO_4, 1 U/mL of β1-3 Galactosidase (NEB, P0726), 5 mM of $MgCl_2$, $CaCl_2$, $CuSO_4$, $NiSO_4$, $MnSO_4$, $FeSO_4$, $CoCl_2$, or $ZnSO_4$ in 50 mM Tris, pH 7.5. Reactions were performed in triplicate and incubated for 1 h at 37 °C. Fluorescence was read at λ_{ex} = 350 nm and λ_{em} = 445 nm using a SpectraMax Plus Microplate Reader (Molecular Devices).

To determine the optimal temperature for N4-ORF16p and E8-ORF27p, 1 μM enzyme and 0.1 mM galactose-3-SO_4 (Toronto Research Chemicals, G155295) were incubated at temperatures ranging from 16 to 70 °C. The optimal pH was determined by incubating 1 μM enzyme and 0.1 mM galactose-3-SO_4, at the respective optimal temperatures in buffers ranging from pH 4.5 to 10.7 (from pH 4.5−5.5, 50 mM sodium acetate; from pH 6.0−7.0, 20 mM sodium phosphate; from pH 7.5−9, 50 mM Tris−HCl; pH 9.7, 20 mM N-cyclohexyl-3-aminopropanesulfonic acid; and pH 10.7, 50 mM carbonate buffer) in 10 μL final reaction volume. Reactions were performed in triplicate. After 1 h incubation, samples were dried in a centrifuge concentrator (Vacufuge plus, Eppendorf) for 30 min. Dried reactions were then labeled with procainamide. The procainamide solution was prepared by dissolving 12 mg of procainamide (Abcam, ab120955) in 110 μL of 70% dimethyl sulfoxide/30% acetic acid solution. This solution and 25 μL water were then transferred to a tube containing 6 mg sodium cyanoborohydride (Sigma, 156159). 20 μL procainamide solution was added to each sample. Reactions were incubated for 1 h at 65 °C. Samples were diluted 1:10 in 100% acetonitrile and separated by UPLC using an ACQUITY UPLC glycan BEH amide 130 Å (2.1 × 150 mm, 1.7 μm) column (Waters, 186004742) on a H-Class ACQUITY instrument (Waters). The injection volume was 3 μL. Solvent A was 50 mM ammonium formate buffer, pH 4.4, and solvent B was 100% acetonitrile. The flow rate was 0.4 mL/min and the gradient used was 0 to 35 min, 12 to 47% solvent A; 35 to 35.5 min, 47 to 70% solvent A; 35.5 to 36.0 min, 70% solvent A; 36 to 36.5 min, 70 to 12% solvent A; 36.5 to 40 min, 12% solvent A. Samples were kept at 4 °C prior to injection and the column

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temperature was 40 °C. Procainamide fluorescence was read at λ_{ex} = 308 nm and λ_{em} = 359 nm, with a data collection rate of 20 Hz.

Specificity of N4-ORF16p and E8-ORF27p was tested on the substrates listed in Table S2 using the optimal reaction conditions identified above (Table S3). Following incubation with the sulfatases, samples were labeled with procainamide and analyzed by UPLC as described above.

Sulfatase Activity on Thyroglobulin *N*-Glycans. *N*-glycans were released from human thyroglobulin (Sigma, T6830) with PNGase F (NEB, P0704). An aliquot containing 100 μg of thyroglobulin was incubated with 1X glycoprotein denaturing buffer and water to make a 100 μL total reaction volume. The glycoprotein was denatured at 100 °C for 10 min. *N*-glycans were cleaved by adding 7 U/mL PNGase F, 1X glycobuffer 2, 1% NP-40, and water for a total reaction volume of 200 μL. The reaction was incubated at 37 °C for 2 h and then cleaned up using a HyperSep Hypercarb SPE column (ThermoFisher, 60106−302). Released *N*-glycans were labeled with 2-aminobenzamide (2-AB; Sigma, 76884), and excess 2-AB was removed through acetone precipitation. To confirm localization of the sulfate to galactose on these *N*-glycans, a sample was incubated with β1−3,4-galactosidase (NEB, P0746), which cannot cleave sulfated galactose. Samples were then analyzed by LC-FLD-MS. High-performance liquid chromatography was carried out on a ThermoFisher Vanquish system using fluorescence detection (FLD) and separated using an ACQUITY UPLC Glycan BEH Amide (130 Å, 1.7 μm, 2.1 mm × 150 mm) column (Waters, 186004742). The mobile phases used were A: 0.1% difluoroacetic acid in water and B: 0.1% difluoroacetic acid in acetonitrile. The column temperature was 45 °C, flow rate 0.3 mL/min, and the gradient was: 70% B to 50% B from 0 to 35 min, 10% B from 35.5 to 38.5 min, with re-equilibration at 70% B from 39 to 45 min. 2-AB fluorescence was detected at λ_{ex} = 320 nm and λ_{em} = 420 nm. LC-MS was performed using a ThermoFisher Orbitrap Eclipse in positive ion mode. Sulfated *N*-glycans were then isolated via HPLC fractionation using the above method and incubated with N4-ORF16p or E8-ORF27p, and activity was reanalyzed by LC-FLD-MS as described above.

RESULTS AND DISCUSSION

Integration of PGM-Targeting Enzymes into the MALDI-MSI Analytical Workflow

The primary aim of this study was to implement new PGM-targeting enzymes in MALDI-MSI to structurally characterize *N*-glycan PGMs *in situ* and visualize their spatial distribution in biological samples. To accomplish this, various PGM-targeting enzymes were incorporated into a common MALDI-MSI *N*-glycan sample preparation and analysis workflow (Figure S2). In this type of analysis, tissues are sectioned, washed, and incubated with the enzyme PNGase F to release *N*-glycans, after which they are typically imaged.[14] In the present study, adjacent tissue sections were first treated with enzymes acting on a specific PGM prior to *N*-glycan release by PNGase F. When the normalized intensity of a mass spectrometry ion decreases after treatment with a PGM-targeting enzyme with known specificity, it is possible to determine several factors: the sugar residue involved, the specific site of the PGM modification on the sugar, and the spatial position of the modified *N*-glycan within the tissue. In principle, different enzyme specificities could be used to discriminate different *N*-glycan modifications, providing a holistic view of the composition and spatial distribution of PGMs *in situ*. In this study, we establish this method and verify the suitability of three PGM-targeting enzyme classes: a phosphodiesterase, a glycosyl hydrolase, and several sulfatases. We conduct this analysis using invertebrate tissue containing phosphorylcholine-modified *N*-glycans, as well as mammalian serum and tissues containing different sulfated *N*-glycans.

GlcNAc-6 Phosphodiesterase Enables Visualization of Phosphorylcholine in Roundworm *N*-Glycans

The roundworm *A. suum* is an intestinal parasitic nematode possessing *N*-glycans modified by the zwitterion phosphorylcholine (PC).[26] PC-modified glycans play an important immunomodulatory role during infections caused by many parasites.[29−35] To confirm the presence of *N*-glycans containing GlcNAc-6-PC and to visualize their presence in *A. suum* sections, a GlcNAc-6 phosphodiesterase (termed GlcNAc-6-PDase) was used in the MALDI-MSI workflow described above.[18] GlcNAc-6-PDase enzyme was previously shown to have remarkable selectivity for GlcNAc-6-PC and no activity on mannose-PC, another common PGM in some invertebrate *N*-glycans, including those from nematodes.[18] When *A. suum* *N*-glycans were released with PNGase F, several *N*-glycans (*m/z* 1301.4510, 1447.5099, and 1669.5844) containing PC were localized either to the cuticle, muscle cell bags, or ovaries, but not in the *Ascaris* intestine or uterus (Figure 1). These data confirm the presence of GlcNAc-6-PC

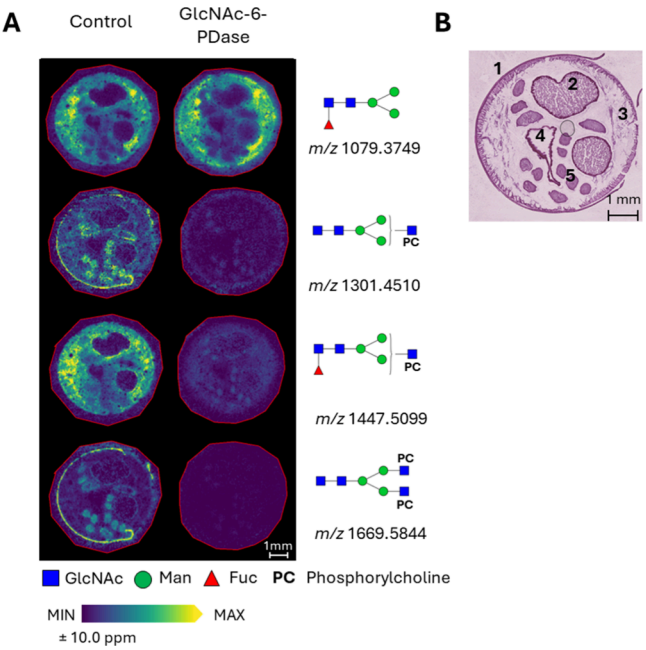


Figure 1. Use of GlcNAc-6-PDase to visualize distribution of phosphorylcholine in *A. suum*. (A) MALDI imaging of neutral and zwitterionic *N*-glycans (Hex3dHex1HexNAc2 + 1Na *m/z* 1079.3749; Hex3HexNAc3 + 1PC + 1Na *m/z* 1301.4510; Hex3dHex1HexNAc3 + 1PC + 1Na *m/z* 1447.5099; Hex3HexNAc4 + 2PC + 1Na *m/z* 1669.5844 ± 10.0 ppm mass error) present in *A. suum*. Serial sections were treated with or without GlcNAc-6-PDase followed by treatment with PNGaseF to release *N*-glycans. *N* = 3. Key: *N*-acetylglucosamine (GlcNAc), mannose (Man), fucose (Fuc), phosphorylcholine (PC). (B) H&E stain of a serial section. Key: 1 − cuticle, 2 − uterus with eggs, 3 − muscle, 4 − intestine, and 5 − ovary.

and illustrate their spatial distribution within *N*-glycans of worm tissue, demonstrating that PGMs can be enzymatically manipulated in a complex *in situ* environment for MALDI-MSI.

Use of Sulfate-Targeting Enzymes to Characterize *N*-Glycans in Human Serum

Sulfation is a common PGM found in nearly all organisms on a wide diversity of glycans, including glycosphingolipids, *N*- and

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O-linked glycans, glycosaminoglycans, and marine exopolysaccharides.[2,36] As such, glycan sulfation influences a variety of processes, including altering the structural stability of surface adhesion, determining the host-specificity in symbiotic relationships, and modulating host−pathogen interactions.[37−43] Accordingly, changes in sulfation patterns have significant effects on glycan function. For instance, decreases in sulfation of dermatan sulfate impede collagen biosynthesis, leading to the congenital connective tissue disease Ehlers-Danlos syndrome.[44] Additionally, increases in *O*-glycan sulfation have been observed in ovarian, colorectal, and breast cancers.[45−48] Changes in glycosaminoglycan (GAG) sulfation have been associated with numerous cancers as well as Alzheimer's disease.[49−53] Sulfated glycans are known for their extensive structural diversity. In structural studies focusing on protein *N*- and *O*-glycosylation, sulfation has been observed on the third, fourth, or sixth carbon of mannose, galactose, GlcNAc, and *N*-acetylgalactosamine (GalNAc).[2] As with other PGMs, glycan sulfation remains largely understudied despite its significant implications. This is due to the nature of sulfation; sulfates are particularly labile and are difficult to characterize by MS, and there is a lack of sensitive complementary techniques besides MS. Thus, to understand the distribution of sulfation *in situ*, it is essential to use enzymes that can both differentiate between various types of sulfated sugars and identify the linkages of sulfate modifications. Ultimately, these enzymes must function efficiently in a highly complex extracellular environment to be suitable for MALDI-MSI.

We established workflow conditions using a sulfate-targeting glycoside hydrolase with model serum glycoproteins. As *N*-glycan sulfation has been identified for some glycoproteins in human serum, we concentrated on this less complex sample in our initial experiments.[6,54] In this experiment, we utilized the enzyme *N*-acetylglucosamine-6-SO_4-hexosaminidase (F10-ORF19p) to enzymatically remove GlcNAc-6-SO_4 reported in serum immunoglobulins, glycoproteins containing *N*-glycans.[2] Nonreactive human serum was spotted onto amine-reactive slides that covalently anchor proteins by their primary amines. Since the *N*-acetylglucosamine-6-SO_4-hexosaminidase specifically targets terminal GlcNAc-6-SO_4 residues, the sample was initially treated with Neuraminidase A and β-galactosidase to remove sialic acid and galactose residues, thereby exposing terminal GlcNAc-6-SO_4 residues. Following this, the samples were incubated with GlcNAc-6-SO_4-hexosaminidase. After incubation, *N*-glycans were released using PNGase F prior to MALDI-MSI (Figure 2 and Figure S3). A marked decrease in signal was noted for several sulfated *N*-glycans, along with a corresponding increase for *N*-glycans where the *N*-acetylglucosamine-6-SO_4-hexosaminidase had removed terminal GlcNAc-SO_4, confirming GlcNAc-6-SO_4 on human serum *N*-glycans and demonstrating the utility of enzymes addressing PGM sulfation.

Discovery and Characterization of New Sulfate-Targeting Enzymes

A more comprehensive arsenal of sulfate-acting enzymes will be needed to systematically decode the sulfated *N*-glycome *in situ*. Although the number of well-characterized sulfate-targeting enzymes has increased in recent years (Table 1), there are still sulfations not targeted by known enzymes. Broad specificity enzymes will allow for a "funnel" approach in characterization, where they are applied first, followed by more specific enzymes to clarify PGMs. The approach would facilitate higher-throughput analysis of sulfated glycans while minimizing sample requirements. However, the discovery of additional sulfatase specificities is needed to effectively characterize all commonly observed sulfate modifications.

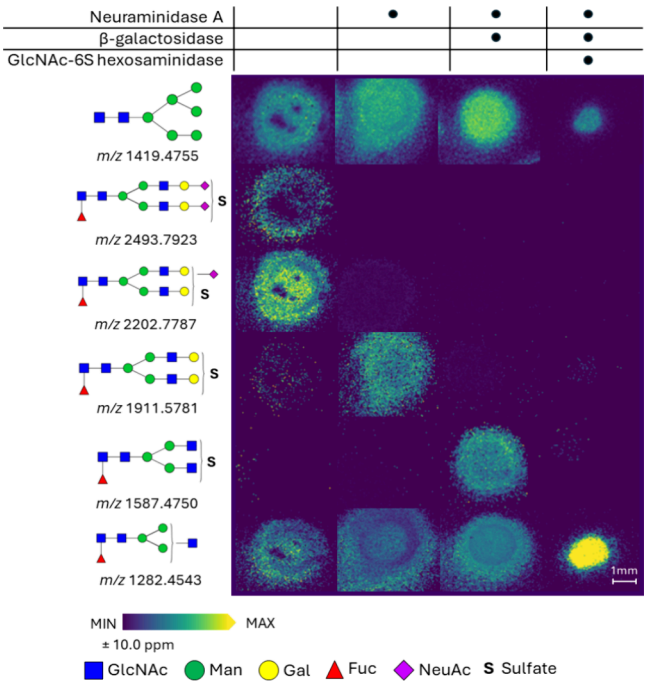


Figure 2. Use of GlcNAc-6-sulfate hexosaminidase to visualize *N*-glycan sulfation in human serum. MALDI imaging of exoglycosidase treated *N*-glycans. (Hex6HexNAc2 *m/z* 1419.4755; Hex5dHex1HexNAc4NeuAc2 + 1SO_4 + 2Na *m/z* 2493.7923; Hex5dHex1HexNAc4NeuAc1 + 1SO_4 + 2Na *m/z* 2202.7787; Hex5dHex1HexNAc4 + 1SO_4 + 2Na *m/z* 1911.5781; Hex3dHex1HexNAc4 + 1SO_4 + 2Na *m/z* 1587.475; Hex3dHex1HexNAc3 *m/z* 1282.4543 ± 10.0 ppm mass error). Serum was spotted on amine-reactive slides, and separate serum spots were treated with neuraminidase A and β-galactosidase to expose GlcNAc residues, followed by treatment with a GlcNAc-6-SO_4 specific hexosaminidase (F10-ORF19p).[2] *N*-glycans were then released using PNGase F. *N* = 2. Key: *N*-acetylglucosamine (GlcNAc), mannose (Man), galactose (Gal), fucose (Fuc), sialic acid (NeuAc), and sulfate (S).

Table 1. Sulfate-Targeting Toolbox[b]

Modification	Endo Activity	Exo Activity
Gal-3-SO_4	BT4683[20a]	BT1636[20a]
Gal-4-SO_4	BT3349[55]; CgsA/B/C[56]	
Gal-6-SO_4		BT3109[20a]
Gal-3 & GalNAc-3-SO_4		BT1622[20]
Gal-4 & GalNAc-4-SO_4		BT3057[20]
GalNAc-4-SO_4	FC509[57]	ARSB[58,59]
Gal-6 & GalNAc-6-SO_4		BT4631[20]
GalNAc-6-SO_4		BT3333[20,60]
GlcNAc-3-SO_4		BT1918[20a]
GlcNAc-6-SO_4		F1-ORF13p[2a]
		GlcNAc-6-S hexosaminidase[2a]

[a]Activity independently verified in-house. [b]Summary of previously characterized enzymes that can be used to study sulfated glycans. A single example of each type of activity has been provided though other enzymes with similar activities have also been characterized, as cited.

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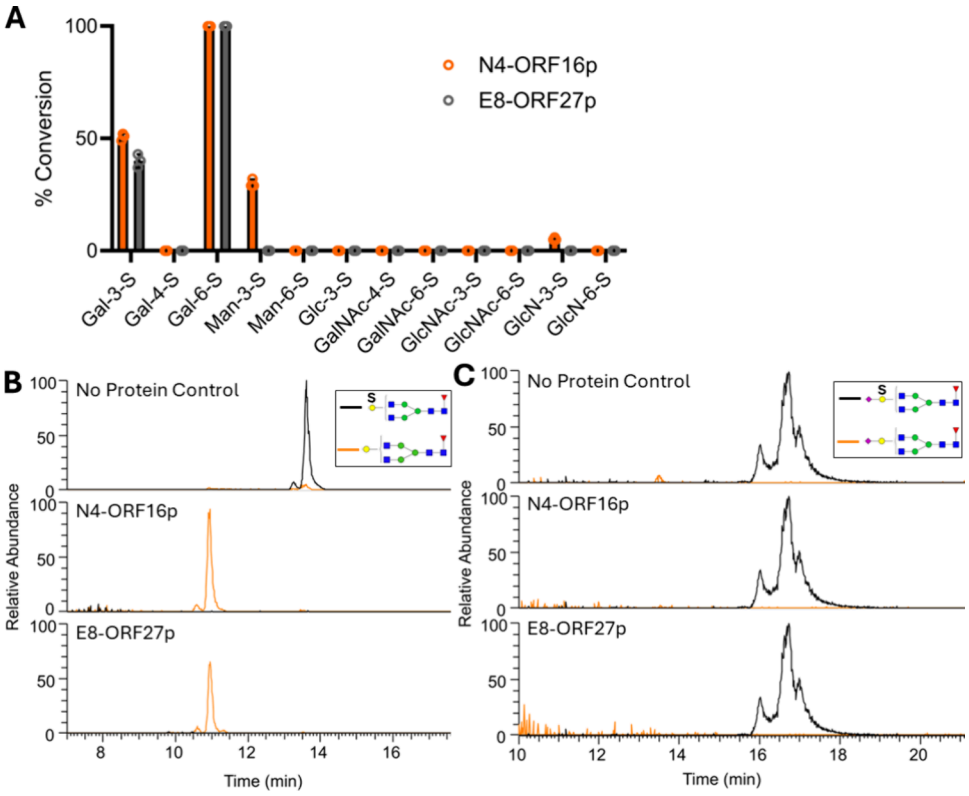


Figure 3. Characterization of galactose-3-SO_4-acting sulfatases identified through functional metagenomic screening. (A) Sulfatase specificity was determined using procainamide labeled sulfated monosaccharides and percent conversion to desulfated form was analyzed by UPLC. Data presented as mean and standard deviation, $N = 3$. (B) Sulfatase activity was evaluated on *N*-glycans isolated from human thyroglobulin, labeled with 2-AB, and treated with β-galactosidase (Figure S7). Isolated *N*-glycans tested contained either a terminal sulfated galactose (Hex4dHex1HexNAc4 + 1SO_4 m/z 913.3220; Hex4dHex1HexNAc4 m/z 873.3428 ± 10.0 ppm mass error) or (C) internal sulfated galactose (Hex4dHex1HexNAc4-NeuAc1 + 1SO_4 m/z 1058.8640; Hex4dHex1HexNAc4NeuAc1 m/z 1018.8900 ± 10.0 ppm mass error). Extracted ion chromatograms are normalized to the sulfated *N*-glycan signal in the no enzyme controls.

To identify new sulfate-targeting enzymes, a functional metagenomic screen was conducted to identify enzymes that act on galactose-3-SO_4. We reasoned that screening for enzymes that act upon sulfated galactose may identify others with broader specificities. The assay used here was adapted from our previous screen, which identified several sulfate-targeting enzymes specific to GlcNAc-6-SO_4. Briefly, 4-methylumbelliferone (4-MU)-labeled galactose-3-SO_4 was incubated with cell lysates from a human gut microbiome fosmid library. When a sulfatase expressed from the fosmid removes sulfate from the substrate, an exogenous β-galactosidase in the assay mixture hydrolyzes 4MU-β-Gal and generates fluorescence (Figure S4A). A clone was considered a "hit" if it produced a signal greater than three standard deviations above the mean background fluorescence. A total of 5,376 clones were screened, resulting in nine positive hits. Upon rescreening with the same assay, five of the clones exhibited a reproducible signal, greater than six standard deviations above background (Figure S4B). Lysates from these five clones were further tested with various substrates and assay conditions: (i) 4MU-Gal-3-SO_4 (without exogenous β-galactosidase), to identify β-galactosidases not inhibited by galactose-3-sulfation, (ii) 4MU-Gal to measure β-galactosidase activity, and (iii) 4MU-SO_4 to assess general sulfatase activity (Figure S4C). Three clones exhibited β-galactosidase activity, supported by the presence of both sulfatase and β-galactosidase genes in those clones. Annotation of the DNA sequences from each of the five candidate fosmid clones revealed overlap in their insert sequences, leading to the identification of two unique sulfatase candidates (Figure S4D). The two candidates, N4-ORF16p and E8-ORF27p, belong to two different sulfatase subfamilies: S1_46 and S1_65, respectively. Sulfatase activity of heterologously expressed and purified candidates was confirmed by incubating each recombinant protein with 4MU-Gal-3-SO_4 in the presence of exogenous β-galactosidase (Figure S5).[36]

Since sulfatases frequently exhibit activity on multiple substrates, the substrate specificities of N4-ORF16p and E8-ORF27p were further evaluated using a panel of sulfated monosaccharides (Table S2). Reactions were carried out using the optimal pH, temperature, and metal cofactors empirically determined for each sulfatase (Table S3 and Figure S6). Activity was quantified as the percent conversion of a sulfated monosaccharide to its desulfated form after 1 h of incubation (Figure 3A). The two sulfatases exhibited overlapping, but distinct specificities. Both enzymes act on galactose-3-SO_4 and galactose-6-SO_4, with an apparent preference for the latter. N4-ORF16p also acted on mannose-3-SO_4, and to a far lesser extent, glucosamine-3-SO_4. These new broad-specificity sulfatases expand our enzymatic toolbox, enabling simultaneous interrogation for two positions, as opposed to sulfatases that act solely on galactose-3-SO_4 (BT1636p) or galactose-6-SO_4 (BT3109p).[20]

N4-ORF16p and E8-ORF27p were then evaluated for their ability to act on more complex sulfated *N*-glycan substrates. The human glycoprotein thyroglobulin is known to contain *N*-

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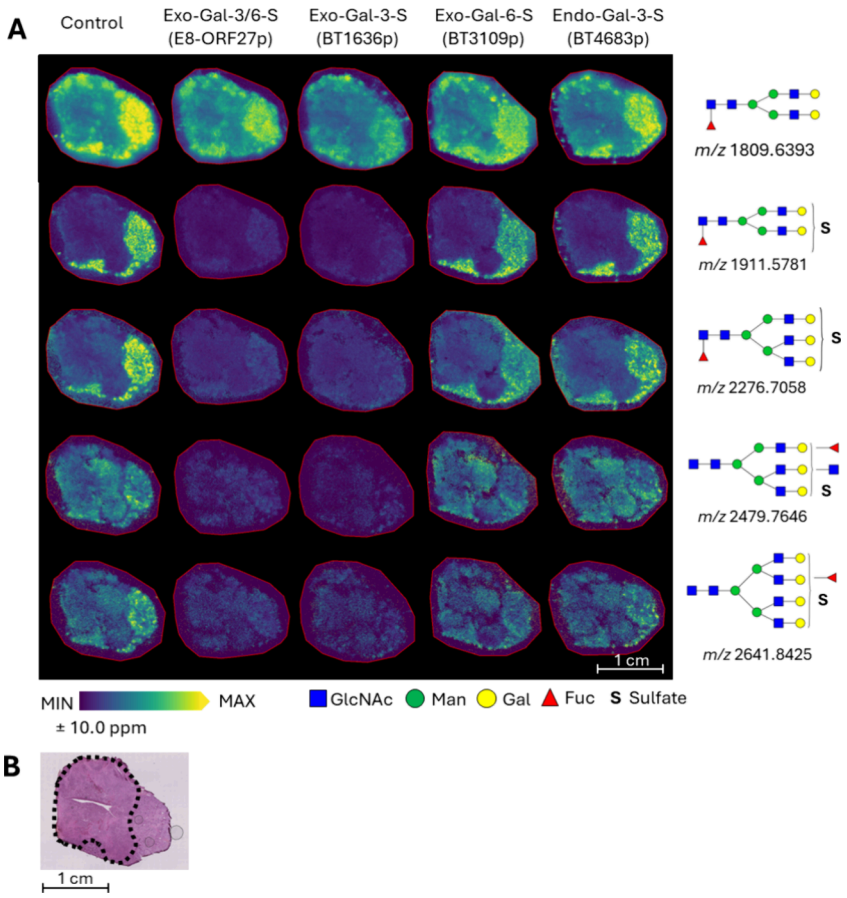


Figure 4. Application of sulfatases to visualize the identity and distribution of sulfated *N*-glycans in cancerous human thyroid tissue. (A) MALDI imaging of sulfated *N*-glycans (Hex5dHex1HexNAc4 + 1Na m/z 1809.6393; Hex5dHex1HexNAc4 + 1SO4 + 2Na m/z 1911.5781; Hex6dHex1HexNAc5 + 1SO4 + 2Na m/z 2276.7058; Hex6dHex1HexNAc6 + 1SO4 + 2Na m/z 2479.7646; Hex7dHex1HexNAc6 + 1SO4 + 2Na m/z 2641.8425 ± 10.0 ppm mass error) present in cancerous human thyroid tissue. Serial sections were treated with either an endogalactose-3/6-sulfatase (E8-ORF27p), an exogalactose-3-sulfatase (BT1636p), an exogalactose-6-sulfatase (BT3109p), or an endogalactose-3-sulfatase (BT4683p) (Luis et al., 2021) followed by PNGase F treatment to release *N*-glycans. N = 2. (B) H&E stain of a neighboring section, cancerous region is outlined in black. Key: *N*-acetylglucosamine (GlcNAc), mannose (Man), galactose (Gal), fucose (Fuc), and sulfate (S).

glycans with sulfated galactose residues.[61] *N*-glycans were released from thyroglobulin, treated with β-galactosidase, and internally or terminally sulfated *N*-glycans were isolated via HPLC fractionation (Figure S7). The resulting sulfated *N*-glycans were then incubated with either N4-ORF16p or E8-ORF27p and analyzed by LC-MS. Both enzymes were effective in removing sulfate from *N*-glycans with a sulfated terminal galactose, but not from glycans with a sulfated internal galactose (Figure 3B,C).

Application of Novel Sulfatases in MALDI-MSI of Cancerous Human Thyroid Tissue

Enzymes that site-specifically remove sulfate from glycans were used in MALDI-MSI to characterize poorly understood sulfated *N*-glycans. Human thyroid glycoproteins contain sulfated galactose, but the position of the sulfate modifications is unknown.[61] In this experiment, parallel sulfatase treatment was performed on serial sections of human thyroid cancer tissue ahead of MSI. The sulfatases used have different specificities, allowing each to discriminate between sulfated positions *in situ*. First, the newly discovered broad specificity sulfatase E8-ORF27p (exogalactose-3/6-sulfatase) was used to determine the nature of galactose sulfation in human thyroid tissue (Figure 4A and Figure S8). A clear decrease in sulfated *N*-glycan structures was observed throughout the tissue, indicating that galactose is modified either on the third or sixth carbon (m/z 1546.4406, 1911.5781, 1954.6072, 2130.657, 2276.7058, 2479.7646, 2495.7895, 2641.8425). A serial section of tissue was then treated with an exogalactose-3-sulfatase (BT1636p)[20] to refine the sulfate position. A decrease in signal was observed for the same sulfated glycans, indicating that all the sulfated *N*-glycans observed in this sample contain galactose-3-SO_4. Additional serial sections were treated with an exogalactose-6-sulfatase (BT3109p)[20] and an endogalactose-3-sulfatase (BT4683p).[20] These two enzymes had no effect, corroborating the observation that terminal galactose-3-SO_4 predominates in this tissue, with no 6-linked galactose sulfation observed. This observation supports a recent study reporting galactose-3-SO_4 sulfation on *N*-glycans extracted from cancerous human thyroid tissue.[25]

These data demonstrate that broadly acting and highly specific PGM-targeting enzymes can be used in tandem in MALDI-MSI workflows to determine the nature and spatial distribution of PGM chemical marks within the *N*-glycome. Furthermore, six of the sulfated *N*-glycans observed in this experiment were found exclusively in the healthy region of the tissue. Although this was a singularly compelling experiment, it underscores the potential significance of spatial PGM information in clinical sample research (Figure 4B).

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Limitations and Perspectives

This study specifically addressed the PGMs of *N*-glycans using well established methods and analysis by MALDI-MSI.[14] Recent work has also demonstrated the feasibility of imaging glycosphingolipids using MALDI-MSI, making them amenable to this type of analysis.[62−65] However, rapid analysis of other glycan classes, such as *O*-glycans, remains limited. Furthermore, glycosaminoglycans and exopolysaccharides are composed of repeating saccharide units and frequently contain multiple PGMs per repeat, significantly complicating their analysis.[66−68] As new methods are developed and the classes of glycans accessible through MSI are expanded, PGM-targeting enzymes can be implemented into any sample preparation workflow. Additionally, PGM-targeting enzymes can be used in traditional exoglycosidase digestion analyses of modified glycans in solution.

To extend the application of PGM visualization to MSI, the toolbox of well-characterized PGM-targeting enzymes will need further expansion. Indeed, many recent studies elucidating novel sulfate-targeting enzymes specificities have established the foundation needed for these experiments (Table 1).[2,20,36,55,56,69−77] However, *N*-glycan sulfation is complex, and the field is still limited by the types of sulfation that can be enzymatically targeted. In addition to sugar and position specificity, most described sulfate-targeting enzymes act only on terminal or internal residues. Since there are no known endoacting GlcNAc-6-SO_4 targeting enzymes, it was necessary for our experiment to first expose the sulfated residues on serum *N*-glycans through a series of enzymatic digestions, increasing the time and reagents required for sample preparation. (Figure 2 and Figure S3). Finally, while sulfatases and other carbohydrate-active enzymes are easily identified using bioinformatics, it is not yet possible to predict their substrate specificity.[78,79] Thus, function-based enzyme screening, such as functional metagenomics, will continue to be a valuable approach for enzyme discovery.[2,18,80]

There are additional opportunities to increase throughput and minimize the amount of sample needed for MSI analysis of PGMs. Running each PGM-targeting enzyme condition separately quickly multiplies the instrument time required, which is not sustainable for high-throughput applications. Fortunately, for homogeneous samples that do not require localization data (e.g., serum samples), multiple enzymatic treatments can be queried on a single slide using a multiwell slide manifold. For samples requiring localization information, it is important to note that this method requires serial tissue sections for each condition tested, which may not be feasible for precious clinical samples. To increase throughput, mixtures of enzymes with differing specificities can be deployed to narrow down the structural possibilities, rather than querying each possible position individually. Faster characterization of glycan modifications will also become possible as more PGM enzymes with broader specificities are described. Alternatively, the use of tissue microarrays will enable screening for disease biomarkers in many samples in parallel, further reducing the instrument time required.[81]

CONCLUSIONS

In this study, we applied a suite of both broad-acting and linkage-specific PGM-targeting enzymes enabling rapid determination of modified glycans, and sugar residue and linkage position profiling, for use in localization studies by MALDI-MSI. Prior work on developing a comprehensive understanding of tissue and disease-specific biomarkers has largely excluded PGMs, as their verification greatly decreases analytical throughput.[16,82−85] The incorporation of PGM-targeting enzymes into MSI studies will increase the accessibility of MSI, enabling more structurally comprehensive *N*-glycomic analyses. Direct visualization of PGM spatial distribution, at linkage-level resolution, will advance the exploration of spatially relevant PGM changes in the *N*-glycome that may correlate with disease.

ASSOCIATED CONTENT

Data Availability Statement

The data sets presented in this study can be found at https://glycopost.glycosmos.org/, GPST000589 and https://metaspace2020.org/project/dreifus-2025. The assembled nucleotide sequences of the 5 fosmid inserts (NEB131-E8, NEB132-J4, NEB133-N4, NEB139-F8, and NEB132-F12) were deposited in GenBank under the accession numbers PV807771-5, respectively.

Supporting Information

The following files are available free of charge. The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jasms.6c00055.

N-glycans released from thyroglobulin with PNGase F, labeled with 2-AB, digested with β-galactosidase, and analyzed by LC/MS (Table S4) (XLSX)

Additional experimental details and data, including additional MSI peaks of interest, enzyme discovery and characterization, as well as substrates and primers used (PDF)

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Author Contributions

J.E.D., C.H.T., and R.R.D. conceived of the study. J.E.D., L.C., E.E.E, G.G., H.B., A.J.H., S.L.F., and V.G. performed experimentation. J.E.D., L.C., E.E.E, and A.J.H. analyzed data. R.J.M. provided the *A. suum* samples. J.E.D. wrote the original draft.; J.E.D., L.C., E.E.E., A.J.H., S.L.F., R.R.D., J.M.F., and C.H.T. reviewed and edited the manuscript. C.H.T, J.M.F., and R.R.D. supervised the project. All authors read and approved the final manuscript.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

PGM,Post-glycosylation modification; MSI,mass spectrometry imaging; LC-MS/MS,liquid chromatography coupled with tandem mass spectrometry; MALDI-TOF,matrix assisted laser desorption ionization − time-of-flight.

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